SARRTHI: Requirement Analysis and Use Cases

Week 4 Deliverable | Reporting period: 27 July to 31 July 2026

Project: SARRTHI, a Proactive Compliance and Payment-Monitoring Platform for MSME Suppliers to Government Transport Bodies

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Document Control

Item	Detail
Document	Requirement Analysis and Use Cases
Version	1.0
Project week	Week 4 of 14 (27 to 31 July 2026)
Prepared by	Project team: Member 1 (Team Lead), Member 2, Member 3, Member 4
Inputs	Project Plan and Market Research Notes (Week 2); Problem Statement and Feasibility Report (Week 3)
Feeds into	SRS first draft (Week 5) and SRS v1.0

1. Introduction

1.1 Purpose

This document analyses the requirements of SARRTHI module by module, defines the actors and business rules, specifies the main use cases, and sets out the priority and phasing matrix. It is the bridge between the research phase and the formal Software Requirements Specification.

1.2 Method

The analysis was divided among the team by module so that each module had a single owner:

Member	Modules analysed
Member 1	Module 1: User, Identity and Access; user classes; assumptions and constraints
Member 2	Modules 2 and 3: Invoice and Payment Monitoring; Compliance Intelligence; the 45-day window and interest rules
Member 3	Modules 4, 5 and 6: Communication and Escalation; Document and Data Capture; Financial Support
Member 4	Modules 7, 8 and 9: Collective Intelligence; Reporting, Admin and Accessibility; Monetisation; non-functional requirements; priority and phasing matrix

1.3 Conventions

- **FR-m.n** is a functional requirement in module m. **UC-nn** is a use case. **BR-n** is a business rule. **NFR-n** is a non-functional requirement.
- **Priority** uses Must, Should and Could. **Phase** uses P1 to P4 from the phasing matrix in Section 8.
- **Academic build** shows whether the 14-week project implements the requirement: Yes, Partial or No (deferred).

2. Stakeholders and Actors

Actor	Description	Main goals
MSME Owner	Primary account holder and decision maker	See what is owed, what is late and what may be claimed; manage profile and settings

Actor	Description	Main goals
MSME Staff	Accountant or office staff with restricted access	Enter orders, invoices, payments and documents
Platform Administrator (future)	Internal team	Verify vendors, moderate data, give support
Trade Body (future)	Industry association	Receive aggregated, anonymised non-compliance data
Lending Partner (future)	Bank or NBFC	Consent-gated view of a vendor's compliance history
MSME ODR Portal (external)	Government dispute channel	Receives a vendor's filing; SARRTHI only prepares information for it
Accounting and GST systems (external, future)	Tally, GST e-invoice	Source of invoice data

3. Business Rules

ID	Rule
BR-1	Payment is due within the agreed period, which may not exceed 45 days from the date of acceptance (MSMED Act, Section 15).
BR-2	The 45-day clock starts from the acceptance date. If no acceptance date is recorded, the invoice date is used, and this assumption is shown to the user.
BR-3	If a written payment term shorter than 45 days is agreed, the shorter term applies.
BR-4	The monitoring window is configurable: number of days, start-date basis, and whether to apply it only to a list of covered buyers. The assumptions behind each result are displayed beside it.

ID	Rule
BR-5	Statutory interest (Section 16) is compound interest with monthly rests at three times the RBI bank rate. The bank rate is a configured value and must be updated when it changes.
BR-6	SARRTHI also offers a simple-interest estimate with a user-set annual rate (default 12%): Principal × Rate × Days overdue ÷ 36,500.
BR-7	Days overdue are counted in whole days from the due date to the as-of date and are never negative.
BR-8	Interest is always labelled an estimate. It is not a guaranteed recovery and not legal advice.
BR-9	Status colours are reserved: green for paid, amber for an upcoming deadline, red for overdue.
BR-10	The procurement quota default is 25% of annual purchases (Public Procurement Policy for MSEs Order, 2012, as amended) and is configurable.
BR-11	Each invoice has an 'MSMED applicable' flag, on by default, so cases where the statutory remedy does not apply can be excluded.
BR-12	A user can read and change only their own records.
BR-13	Money is stored and calculated as exact decimals, never as floating-point numbers.
BR-14	The disclaimer 'Academic Project — Not an Official Government Portal.' is displayed on every screen. No official logos are used.

Worked example

Invoice of ₹10,00,000, fully unpaid, 20 days past the due date.

Method	Calculation	Result
Simple estimate at 12% (BR-6)	10,00,000 × 12 × 20 ÷ 36,500	₹6,575.34

Method	Calculation	Result
Statutory estimate at 20.25% (3 × 6.75%, illustrative)	10,00,000 × (1 + 0.2025 × 20 ÷ 365) − 10,00,000	₹11,095.89
Statutory estimate, 75 days overdue	2 monthly rests plus 15 days at 20.25%	₹42,639.92

The 6.75% bank rate above is an illustrative configuration value, not a statement of the current RBI rate.

4. Module-wise Requirement Analysis

4.1 Module 1: User, Identity and Access

ID	Requirement	Priority	Phase	Academic build
FR-1.1	Register with email, password, name and company	Must	P1	Yes
FR-1.2	Log in and out with stateless token (JWT) authentication; query current session	Must	P1	Yes
FR-1.3	Maintain a business profile with Udyam, GSTIN and PAN	Must	P1	Yes (stored, not verified)
FR-1.4	Verify the Udyam registration against the official source	Should	P1	No
FR-1.5	Role-based access for Owner and Staff	Should	P1	Partial
FR-1.6	Isolate each user's data from every other user	Must	P1	Yes
FR-1.7	Change password	Should	P1	Yes
FR-1.8	Record consent and privacy preferences	Must	P1	Partial

4.2 Module 2: Invoice and Payment Monitoring (Core)

ID	Requirement	Priority	Phase	Academic build
FR-2.1	Create, view, edit and delete invoices with number, buyer, amount, dates, terms and linked order	Must	P1	Yes
FR-2.2	Show due date, days remaining or overdue, and status for every invoice	Must	P1	Yes
FR-2.3	Record part and full payments and recalculate the balance	Must	P1	Yes
FR-2.4	Configurable 45-day monitoring with displayed assumptions	Must	P1	Yes
FR-2.5	Interest estimate per overdue invoice (BR-5 and BR-6)	Must	P1	Yes
FR-2.6	Interest calculator for what-if amounts and rates	Should	P1	Yes
FR-2.7	Search, filter and export the invoice list	Should	P1	Yes
FR-2.8	Create, edit, view and delete purchase orders with delivery and acceptance tracking	Must	P1	Yes
FR-2.9	Document-completeness check for filing readiness	Should	P1	Partial (evidence checklist)
FR-2.10	Working-capital gap estimate from pending invoices	Should	P1	Yes
FR-2.11	Contract-clause checker that flags a missing written payment term	Could	P1	No

FR-2.8 was added during analysis because the acceptance date recorded on an order starts the payment clock.

4.3 Module 3: Compliance Intelligence

ID	Requirement	Priority	Phase	Academic build
FR-3.1	Quota dashboard showing the estimated MSME procurement share against the 25% target	Should	P2	Yes (indicative)
FR-3.2	Compliance checklist the user can tick off	Should	P2	Yes
FR-3.3	Buyer risk profile: overdue invoices and average delay by buyer	Should	P2	Yes
FR-3.4	Rule-based pre-bid risk score	Should	P2	Yes
FR-3.5	Track tenders with MSME eligibility and saved tenders	Should	P2	Yes
FR-3.6	Specification-change alert feed	Could	P2	No
FR-3.7	Buyer-rating feedback after contract closure	Could	P2	No

FR-3.5 was added because the pre-bid risk score is shown at the point of bidding.

4.4 Module 4: Communication and Escalation

ID	Requirement	Priority	Phase	Academic build
FR-4.1	In-app alerts for upcoming and overdue deadlines	Must	P1	Yes
FR-4.2	WhatsApp reminders at day 30 and day 40	Should	P2	No
FR-4.3	Escalation timeline: reminder, formal notice, ODR-ready package	Should	P2	Partial (timeline guidance)
FR-4.4	ODR pre-fill from tracked records	Could	P2	No

ID	Requirement	Priority	Phase	Academic build
FR-4.5	Vernacular rights chatbot and full multi-language interface	Could	P2	No

4.5 Module 5: Document and Data Capture

ID	Requirement	Priority	Phase	Academic build
FR-5.1	Document vault: upload, list, download and delete, with categories, linked to orders and invoices	Must	P1	Yes
FR-5.2	Evidence checklist for each order	Should	P1	Yes
FR-5.3	OCR capture from a phone camera	Could	P1	No
FR-5.4	Tally and GST e-invoice import	Could	P2	No

4.6 Module 6: Financial Support

ID	Requirement	Priority	Phase	Academic build
FR-6.1	Export invoices, orders and buyer delays to CSV	Should	P2	Yes
FR-6.2	Print-friendly reports; PDF and Excel export	Could	P2	Partial (print view)
FR-6.3	TReDS referral once an invoice is overdue past a threshold	Could	P2	No

4.7 Module 7: Collective Intelligence

Crowd-sourced quota tracker, specification-change alert network, collective escalation bundling and trade-body handoff. All are **Could** priority in **P3**, and none is in the academic build. They need a meaningful number of vendors before the data is reliable. The quota

dashboard (FR-3.1) and the alert feed (FR-3.6) are the single-vendor versions of the first two, so one underlying service is built for each pair.

4.8 Module 8: Reporting, Admin and Accessibility

ID	Requirement	Priority	Phase	Academic build
FR-8.1	Dashboard with KPIs, monthly financials, invoice status, alerts, buyer risk and activity feed	Must	P1	Yes
FR-8.2	Responsive layout and accessible interface	Must	P1	Yes
FR-8.3	Passwords stored as hashes; traffic protected by TLS in deployment	Must	P1	Partial
FR-8.4	Audit trail of significant actions	Should	P1	Partial (activity feed)
FR-8.5	Demo mode with realistic sample data and a reset function	Should	P1	Yes
FR-8.6	Admin dashboard, offline-first mode	Could	P1	No

4.9 Module 9: Monetisation

Freemium tier management and a consent-gated lender API. **Could** priority, **P3** and **P4**, not in the academic build.

5. Use Case Analysis

5.1 Use case list

ID	Use case	Actors	Requirements
UC-01	Register an account	Owner	FR-1.1
UC-02	Log in and log out	Owner, Staff	FR-1.2

ID	Use case	Actors	Requirements
UC-03	Maintain business profile	Owner	FR-1.3
UC-04	Manage purchase orders and record acceptance	Owner, Staff	FR-2.8
UC-05	Record an invoice	Owner, Staff	FR-2.1
UC-06	View payment clock and status	Owner, Staff	FR-2.2
UC-07	Record a payment	Owner, Staff	FR-2.3
UC-08	View interest estimate and use calculator	Owner	FR-2.5, FR-2.6
UC-09	Configure 45-day monitoring	Owner	FR-2.4
UC-10	Track and save tenders	Owner	FR-3.5
UC-11	Review compliance summary, checklist and buyer risk	Owner	FR-3.1 to FR-3.4
UC-12	Manage documents and evidence	Owner, Staff	FR-5.1, FR-5.2
UC-13	View dashboard and alerts	Owner	FR-8.1, FR-4.1
UC-14	Generate and export reports	Owner	FR-6.1, FR-2.10
UC-15	Change settings and password	Owner	FR-1.7, FR-1.8
UC-16	Load or reset demo data	Owner	FR-8.5

5.2 Detailed use cases

UC-05: Record an invoice

Field	Detail
Actor	MSME Owner or Staff

Field	Detail
Precondition	The user is logged in.
Main flow	1. The user opens Invoices and chooses Add Invoice. 2. The user enters invoice number, buyer, amount, invoice date, acceptance date (optional), payment terms and linked order (optional). 3. The system validates the input. 4. The system calculates the due date under BR-1 to BR-4. 5. The system saves the invoice and adds an entry to the activity feed. 6. The invoice appears in the list with its status.
Alternate flows	A. A required field is missing or the amount is not positive: the system shows an inline error and does not save. B. No acceptance date: the system uses the invoice date and shows the assumption.
Postcondition	The invoice is stored for this user only and the dashboard figures are refreshed.

UC-06: View payment clock and status

Field	Detail
Actor	MSME Owner or Staff
Precondition	At least one invoice exists.
Main flow	1. The user opens an invoice. 2. The system shows the due date and the days remaining or days overdue. 3. The system shows the status in the reserved colour (BR-9). 4. The system shows the monitoring assumptions (BR-4).
Alternate flows	A. Invoice fully paid: status is paid and no clock is shown. B. The MSMED flag is off: the system shows the agreed due date only.
Postcondition	None; the use case only reads data.

UC-07: Record a payment

Field	Detail
Actor	MSME Owner or Staff
Precondition	The invoice has an outstanding balance.
Main flow	1. The user chooses Record Payment. 2. The user enters amount, date, method and reference. 3. The system checks that the amount is positive and not more than the balance. 4. The system saves the payment and reduces the balance. 5. The system updates the status and refreshes the dashboard, reports and buyer delays.
Alternate flows	A. Amount exceeds the balance: the system rejects it with a message. B. Balance reaches zero: the invoice becomes paid.
Postcondition	Balance, status and all dependent figures are consistent.

UC-08: View interest estimate

Field	Detail
Actor	MSME Owner
Precondition	The invoice is overdue and has an outstanding balance.
Main flow	1. The user opens the invoice and views the interest panel. 2. The system calculates the estimate under BR-5 and BR-6 as at today's date. 3. The system shows the amount, the method and the label 'Estimate only'.
Alternate flows	A. The invoice is not overdue: the system shows zero interest. B. The user changes the rate in the calculator: the system recalculates without saving.
Postcondition	None.

UC-11: Review compliance and buyer risk

Field	Detail
Actor	MSME Owner
Precondition	The user is logged in.
Main flow	1. The user opens Compliance. 2. The system shows the quota summary against the 25% target. 3. The system shows the checklist and the risk score for each buyer. 4. The user ticks completed checklist items and the system saves them.
Alternate flows	A. No buyer history exists: the system shows an empty state that explains how scores are produced.
Postcondition	Checklist progress is stored.

6. Preliminary Data Model

Entity	Purpose	Key attributes
User	Account and role	email, password hash, full name, company, role
BusinessProfile	Vendor identity	Udyam number, GSTIN, PAN, category, products, contact
PurchaseOrder	Order and delivery	order number, buyer, value, dates, status, proof of delivery
Invoice	Receivable	number, buyer, amount, amount paid, balance, dates, terms, due date, status, MSMED flag
Payment	Receipt against an invoice	amount, date, method, reference
Tender	Opportunity	reference, title, buyer, value, deadline, MSME eligibility, saved
Document	Evidence	name, category, type, size, linked order or invoice

Entity	Purpose	Key attributes
ComplianceChecklistItem	Compliance task	text, category, completed
BuyerRisk	Buyer reliability	overdue invoices, average delay days, score
Activity	Activity feed and audit	kind, message, timestamp
UserSettings	Preferences	notification and privacy options, bank details

Every record except User belongs to exactly one user. An invoice may link to one order. A payment belongs to one invoice. A document may link to an order and an invoice.

7. Non-Functional Requirements

ID	Category	Requirement
NFR-1	Performance	95% of API requests answered in under 2 seconds under normal load
NFR-2	Security	Passwords hashed; JWT authentication; access checked on every request; TLS in transit
NFR-3	Data accuracy	Money held as exact decimals; calculations covered by unit tests
NFR-4	Usability	Core tasks, such as adding an invoice or viewing its status, within 3 actions of login
NFR-5	Accessibility	Responsive layout, readable contrast, keyboard operation, meaningful labels
NFR-6	Availability	Target 99.5% uptime for tracking and alerting in a production deployment
NFR-7	Maintainability	Single-language Java backend, layered structure, consistent style enforced by tooling
NFR-8	Portability	Containerised deployment; relational database that can move between providers

ID	Category	Requirement
NFR-9	Compliance	Follow the DPDP Act, 2023: consent, minimisation, deletion and withdrawal
NFR-10	Transparency	Every estimate shows its assumptions and a non-legal-advice label

8. Priority and Phasing Matrix

Phase	Content	Rationale
P1: MVP	Module 1; Module 2 including orders; document vault and evidence checklist; dashboard, security basics, demo data	Needs only vendor-supplied data and has no dependency on uncertain integrations
P2: Core expansion	Module 3; Module 4 (alerts, escalation); CSV and report exports; Tally and GST import; TReDS referral	Builds the full compliance and escalation experience once the tracker is proven
P3: Growth	Module 7 collective intelligence; Module 9 freemium tiers	Needs a meaningful vendor base
P4: Advanced	AI default prediction, voice assistant, blockchain document proof, native apps, lender API	Pursued after product-market fit

The 14-week academic build covers all of P1 and the practical parts of P2 (compliance summary, buyer risk, tender tracking, exports). It leaves out OCR, WhatsApp, third-party integrations, collective intelligence and monetisation.

9. Assumptions and Constraints

Assumptions

- Vendors will enter their own orders, invoices and payments; the system cannot work without this minimum input.
- The acceptance date is known to the vendor and can be recorded.

- Interest figures are indicative; the vendor or an adviser verifies them before any claim.

Constraints

- The backend is implemented in Java with Spring Boot.
- No public IREPS API exists, so government data is not a dependency.
- SARRTHI performs no legal, arbitration or dispute-resolution function.
- The project must be delivered within 14 weeks by a team of four.

Appendix A: Traceability to the Final Implementation

Added at final submission to show where each module is built in the Java backend.

Module	Spring Boot controller and endpoints	Service and utility classes
1 Identity and access	AuthController (/api/auth), BusinessController (/api/businesses/profile), SettingsController (/api/settings)	AuthService, BusinessProfileService, SettingsService, SecurityConfig, JwtTokenProvider, JwtAuthenticationFilter
2 Invoices and orders	InvoiceController (/api/invoices, /api/payments, interest-estimate), OrderController (/api/orders)	InvoiceService, PaymentService, PurchaseOrderService, MsmedCalculator
3 Compliance and tenders	ComplianceController (/api/compliance), TenderController (/api/tenders)	ComplianceService, TenderService
5 Documents	DocumentController (/api/documents, upload, download, evidence-checklist)	DocumentService
6 and 8 Reports and dashboard	DashboardController (/api/dashboard, /api/activity, /api/reports)	DashboardService, ReportService, ActivityService
Demo data	DemoController (/api/demo/seed)	DemoSeedService

The backend has 20 automated test methods across five test classes, including seven for the MSMED calculator.